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Predicting Type 2 Diabetes Using Logistic Regression and Machine Learning Approaches

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Abstract

Diabetes mellitus is one of the most common human diseases worldwide and may cause several health-related complications. It is responsible for considerable morbidity, mortality, and economic loss. A timely diagnosis and prediction of this disease could provide patients with an opportunity to take the appropriate preventive and treatment strategies. To improve the understanding of risk factors, we predict type 2 diabetes for Pima Indian women utilizing a logistic regression model and decision tree—a machine learning algorithm. Our analysis finds five main predictors of type 2 diabetes: glucose, pregnancy, body mass index (BMI), diabetes pedigree function, and age. We further explore a classification tree to complement and validate our analysis. The six-fold classification tree indicates glucose, BMI, and age are important factors, while the ten-node tree implies glucose, BMI, pregnancy, diabetes pedigree function, and age as the significant predictors. Our preferred specification yields a prediction accuracy of 78.26% and a cross-validation error rate of 21.74%. We argue that our model can be applied to make a reasonable prediction of type 2 diabetes, and could potentially be used to complement existing preventive measures to curb the incidence of diabetes and reduce associated costs.

Keywords: decision tree; diabetes risk factors; machine learning; prediction accuracy.

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Figures



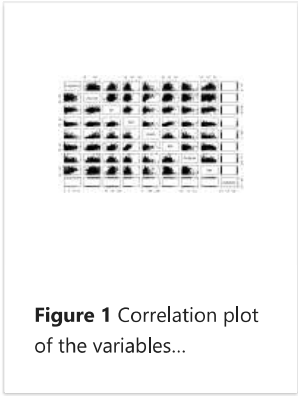


Figure 1 Correlation plot of the variables...

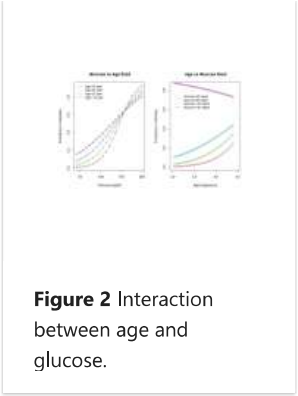


Figure 2 Interaction between age and glucose.

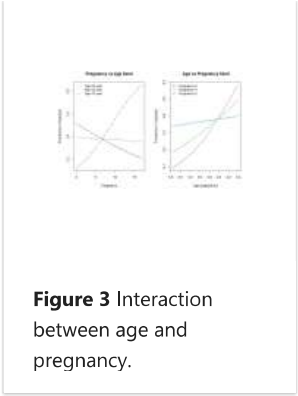


Figure 3 Interaction between age and pregnancy.

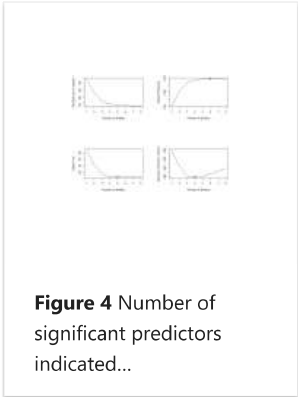


Figure 4 Number of significant predictors indicated...

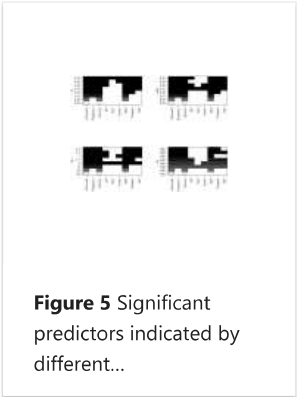


Figure 5 Significant predictors indicated by different...

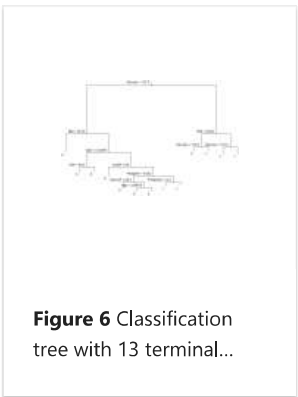


Figure 6 Classification tree with 13 terminal...

All figures (8)

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